Krapa Sandeep

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.53
Sri Chaitanya Junior College	Class XII (Board Of Intermediate Education Andhra Pradesh)	2022	96.50%
Dr KKR's Gowtham E/M High School	Class X (Board of Secondary Education Andhra Pradesh)	2020	98.33%

ACADEMIC ROLES

Teaching Assistant - Database Management Systems | Prof. Anupam Sanghi

Teaching Assistant - Data Structures and Applications | Prof. Subrahmanyam Kalyanasundaram

PROJECTS

WealthPilot - SME Loan Underwriting & Credit Research Assistant | (Coursework / Prof. Praveen Tammana

- Developed a **multi-agent AI assistant** for SME loan underwriting and credit research, using **financial-document parsing, risk-analysis tools, LLM APIs, and persistent memory**.
- Built a **policy-aware RAG system** with vector search, **LangGraph** for agent orchestration, and **MCP** for tool integration to support credit assessment and loan recommendations.
- Added **observability and reliability checks**, along with **bias/compliance evaluation and human approval**, to make the AI-driven decisions safer and more explainable.

Observability Platform for Logs and Exceptions

- Engineered a **real-time telemetry pipeline for logs, metrics, and traces** using **Envoy, Vector, Redpanda, Flink, and ClickHouse.**
- Designed **hot/cold data pipelines** with **intelligent routing, tail sampling, and error-rate aggregation** for efficient telemetry processing.
- Implemented **ML-driven observability** using **ClickHouse and Milvus** for analytical storage and semantic retrieval, supporting **anomaly detection, root-cause classification, and latency forecasting**.
- Developed real-time dashboards using **Grafana, Prometheus, FastAPI, and Next.js**, with **PII detection and masking** for sensitive telemetry.

Multi-Client Quiz System with LLM Integration (Coursework / Prof. Kotaro Kataoka)

- Engineered a concurrent **C++ client-server quiz system using AF_INET TCP sockets, pthreads, and mutexes**, supporting multi-client answer evaluation and scoring.
- Designed a custom **application-layer protocol with KEEP_ALIVE/ALIVE_OK heartbeats** and **SQLite-backed** score and leaderboard persistence.
- Implemented a **file-based LLM cache with expiry and access tracking**, reducing redundant requests for repeated quiz generation.
- **Integrated FastAPI/Groq with React/Vite via a WebSocket-TCP bridge** and analyzed **latency and throughput using Wireshark**.

Inventory Management System (Coursework / Prof. Shirshendu Das)

- Designed a **BCNF-oriented MySQL schema with 10+ relations and 7K+ tuples**, using foreign keys, constraints, and **multi-warehouse inventory modeling**.
- Implemented **triggers, stored procedures, transactions, and row-level locking** for automatic stock updates and **race condition safe** inventory operations.
- Built a **Flask + React full-stack application with REST APIs and RBAC** for Admin, Staff, Viewer, and Auditor roles, supporting inventory, sales, purchases, and reporting.

RISC-V Assembler & Cache Simulator(Coursework / Prof. Rajesh Kedia)

- Developed a **C++ RISC-V assembler** supporting (**R, I, S, B, J, U) formats**, converting instructions to hexadecimal machine code.
- Implemented a **RISC-V instruction simulator** supporting register, memory, arithmetic, logical, branch, and jump operations.
- Built a **cache simulator** with **direct-mapped/set-associative caches**, **LRU/FIFO/RANDOM replacement**, and hit/miss analysis.

SKILLS

- **Programming:** C, C++, Python, SQL
- **Web & Backend:** ReactJS, FastAPI, Flask, Next.js, REST APIs, WebSockets
- **Databases & Systems:** MySQL, SQLite, Linux, TCP/IP, POSIX Threads, Git

RELEVANT COURSES

- **Core Courses:** Software Engineering, Data Structures, Algorithms, DBMS, Operating Systems, Computer Architecture, Theory of Computation, FOML , Computer Networks, Deep Learning, Fraud Analytics, Compilers,Multi Agent Systems.
- **Math Courses:** Linear Algebra, Probability and Random Variables, Transform Techniques, Calculus, Statistics.

SCHOLASTIC ACHIEVEMENTS

- **CodeForces:** Attained a competitive rating of **1167** and achieved a peak global rank of **2,024** in Round 1120 (Div. 2).

- **Leetcode:** Solved 250+ dsa problems.

EXTRACURRICULAR

- Hospitality Coordinator | Milan Inter-Hostel Competition
- Active volunteer of NSS, IITH (National Service Scheme)